

Build the Coupled-Coil Instrument

A low-voltage pulse and resonance platform

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

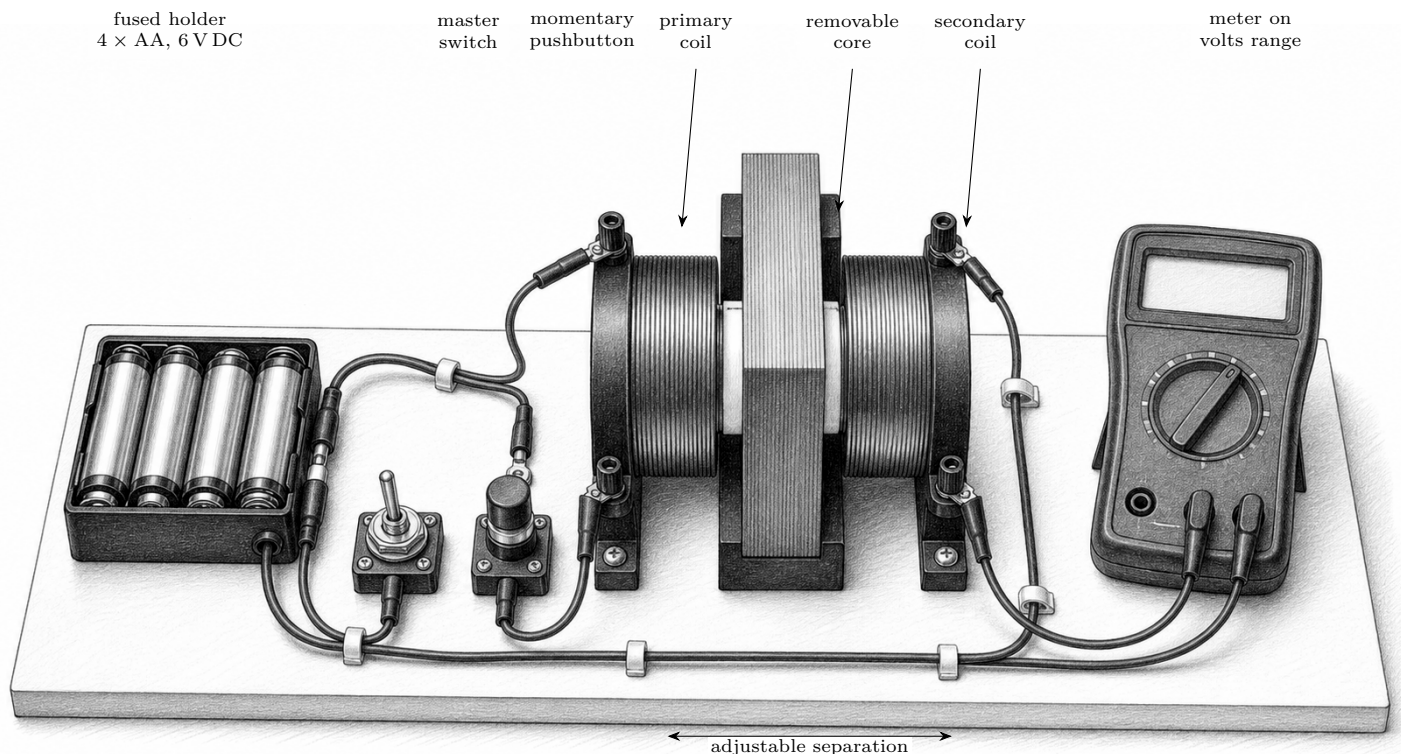
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- 2–4 AA cells in fused, switched holder
- Two insulated coils on a removable ferrite or laminated-iron core
- Pushbutton
- LED with 1 k Ω resistor
- Multimeter
- Cardboard base and cable ties

Predict first

Which change should most increase the receiver indication: aligning the coils, adding the core, or holding DC continuously?



Investigate

1. Mount components so no bare lead can bridge the battery. Provide strain relief and a clearly labeled master switch.
2. Connect the primary through the pushbutton. Connect the secondary only to its meter or correctly resistor-limited LED.
3. Adult performs continuity, resistance, polarity, meter-jack, and short-path inspection while deenergized.
4. Pulse for less than one second; record receiver response. Compare coil gap, alignment, turns, and core one at a time.
5. Disconnect the battery for every adjustment. Stop on warmth or unexpected behavior.

Explain from the evidence

This instrument combines current-produced field, changing flux, mutual induction, and geometry. It deliberately does not make sparks and is not a Tesla coil. Continuous DC gives little secondary response after the switching transient. A measured low-energy pulse provides more useful evidence than an uncontrolled high-voltage spectacle.

Change one thing

Create a calibration chart of receiver peak versus separation, with three trials and uncertainty bars. Photograph or redraw the final wiring and label every energy-transfer stage.

Evidence record	
Prediction	_____
One change	_____
Observation	_____
Claim + because	_____

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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